AI DATA ANALYTICS MINI PROJECT 1

EXPLORATORY DATA ANALYSIS (EDA)

Group 3, Member: Yam Ching Long

Title

- Video Game Sales (1980 – 2024) Analysis

Objectives

- Identify key factors which influence sales performance (e.g. platform, genre, region)
- Provide sales strategy for game publishers and developers

Dataset

- A collection of 64,000+ video games across >40 years of gaming history
- Contains 14 features

img	title	console	genre	publisher	developer	critic_score	total_sales	na_sales	jp_sales	pal_sales	other_sales	release_date	last_update
tccc.jpg	Grand Theft Auto V	PS3	Action	Rockstar Games	Rockstar North	9.4	20.32	6.37	0.99	9.85	3.12	17-09-2013	NaN
tccc.jpg	Grand Theft Auto V	PS4	Action	Rockstar Games	Rockstar North	9.7	19.39	6.06	0.60	9.71	3.02	18-11-2014	03-01-2018
	Grand												

Dataset Characteristics

Variable Name	Description	Variable Type	Category	Data Type
img	URL for game cover	Qualitative	Nominal	Object
title	Game title	Qualitative	Nominal	Object
console	Platform that the game was released on	Qualitative	Nominal	Object
genre	Genre of the game	Qualitative	Nominal	Object
publisher	Publisher of the game	Qualitative	Nominal	Object
developer	Developer of the game	Qualitative	Nominal	Object
critic_score	Metacritic score (out of 10)	Quantitative	Continuous	Float64
total_sales	Global sales of copies in millions	Quantitative	Continuous	Float64
na_sales	North American sales of copies in millions	Quantitative	Continuous	Float64
jp_sales	Japanese sales of copies in millions	Quantitative	Continuous	Float64
pal_sales	European & African sales of copies in millions	Quantitative	Continuous	Float64
other_sales	Other regional sales of copies in millions	Quantitative	Continuous	Float64
release_date	Date that the game was released on	Temporal	N/A	Object
last_update	Date that the data was last updated	Temporal	N/A	Object

Data Preprocessing

- Check the dimension of the dataset

```
(64016, 14)
```

- Drop “img” and “last_update” (irrelevant)
- In “developer”, fill the missing values with “Unknown” (retain valuable data)

Data Preprocessing

- In “critic_score”, fill the missing values with KNN Imputation
  - Find the most similar K neighbors by calculating the distance between data
  - Fill the missing values by mean or median of the neighbors
  - Here K=5

```python
from sklearn.impute import KNNImputer
# Select columns
columns_to_use = ['critic_score', 'total_sales', 'na_sales', 'jp_sales', 'pal_sales', 'other_sales']
df_subset = df[columns_to_use].copy()

# Apply KNN imputer
imputer = KNNImputer(n_neighbors=5)
df_imputed = pd.DataFrame(
    imputer.fit_transform(df_subset),
    columns=df_subset.columns
)

# Update critic_score
df['critic_score'] = df_imputed['critic_score']
```

Data Preprocessing

- In “total_sales”, delete the rows with value 0 and with missing values
  - 0: impossible
  - Missing values: also have missing values in regional sales -> delete directly

H	I	J	K	L
total_sales	na_sales	jp_sales	pal_sales	other_sales

Data Preprocessing

- In “na_sales”, “jp_sales”, “pal_sales” and “other_sales”, fill the missing values with 0
  - Regional sales can be 0 (Not released in that region or prohibited from selling)
- In “other_sales”, delete erroneous input (date)

Data Preprocessing

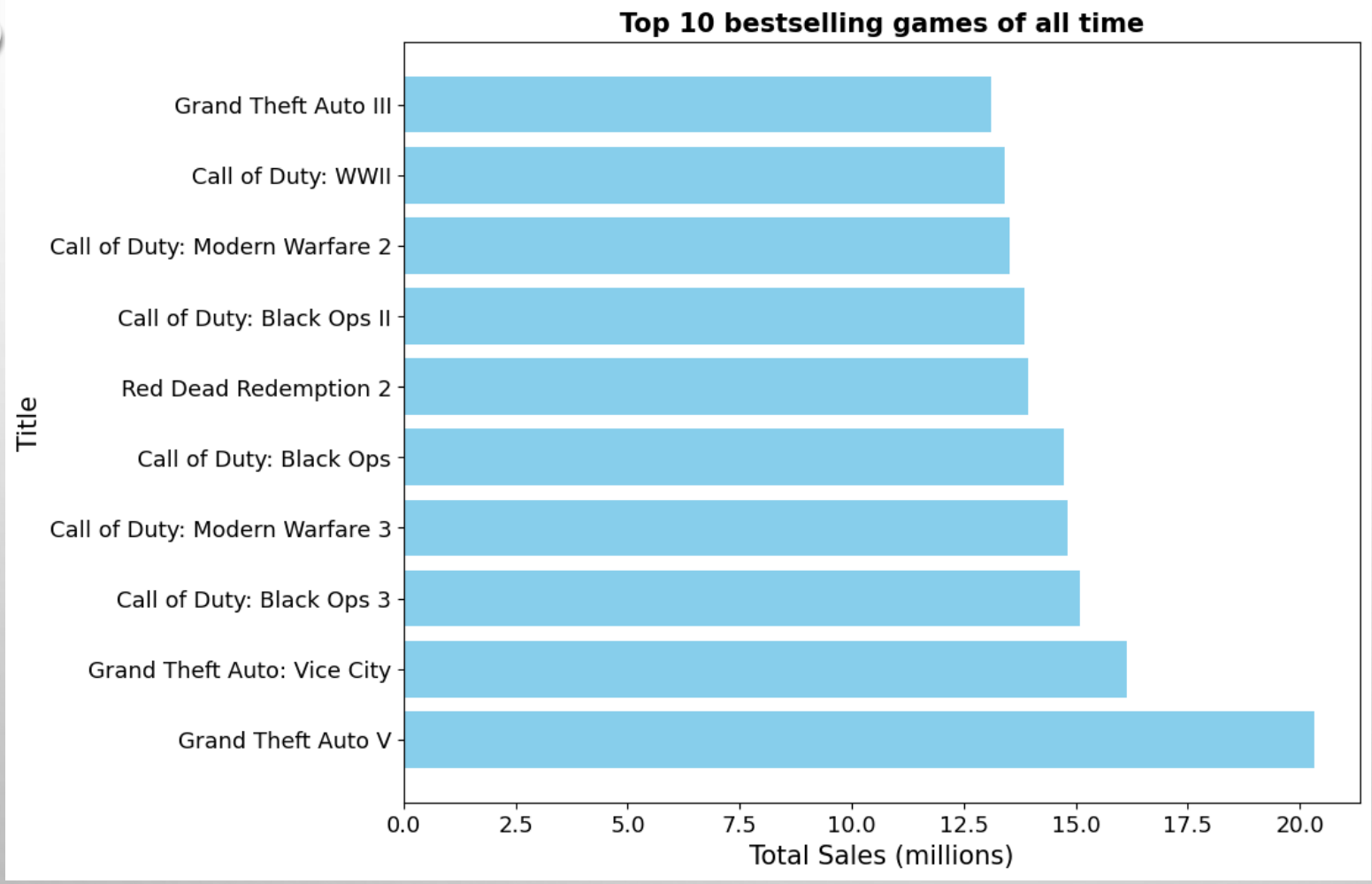
- In ‘release_date’, drop the rows with missing values (date cannot be estimated)
- Only year is extracted (month and day are not important)

Data cleansing is completed!

title	console	genre	publisher	developer	critic_score	total_sales	na_sales	jp_sales	pal_sales	other_sales	release_year
Grand Theft Auto V	PS3	Action	Rockstar Games	Rockstar North	9.4	20.32	6.37	0.99	9.85	3.12	2013
Grand Theft Auto V	PS4	Action	Rockstar Games	Rockstar North	9.7	19.39	6.06	0.60	9.71	3.02	2014

Dimension: `(17508, 12)`

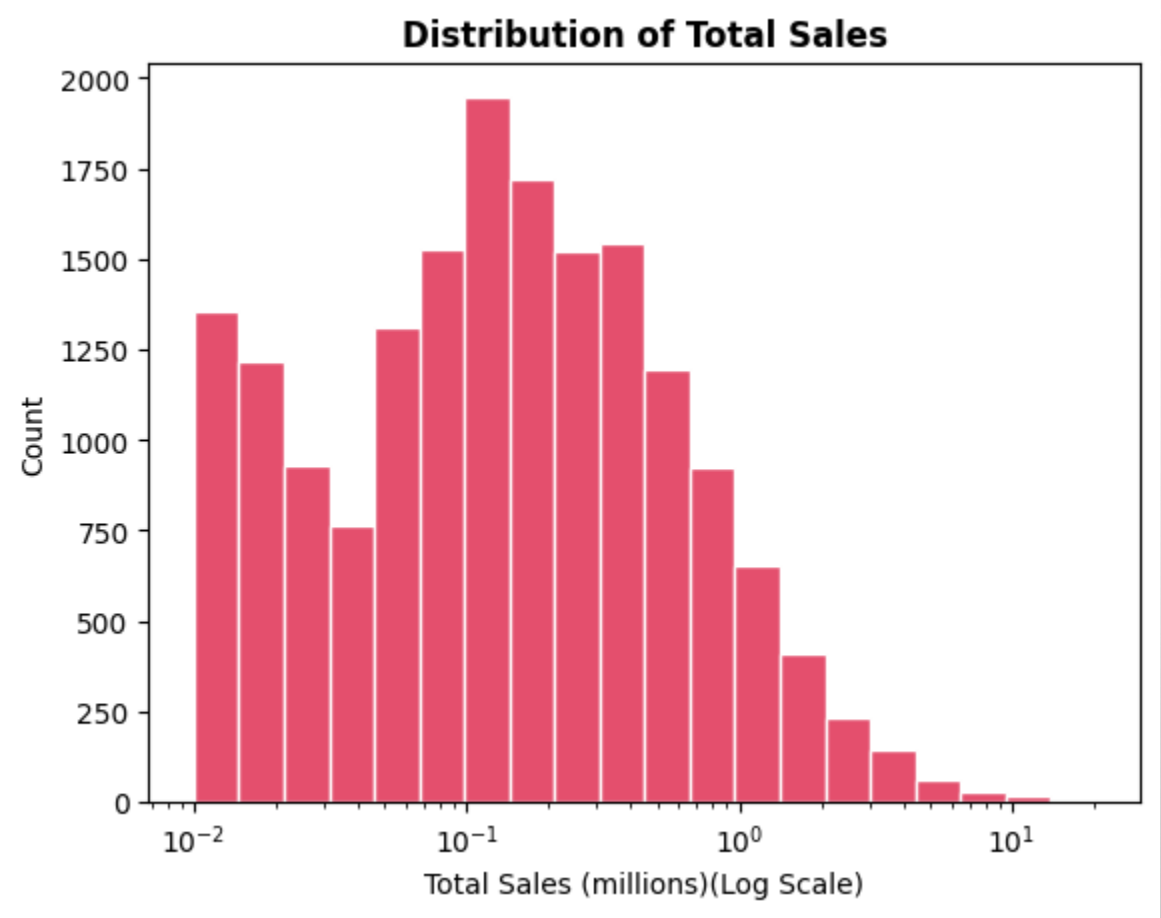
We first take a look at the top 10 bestselling games…

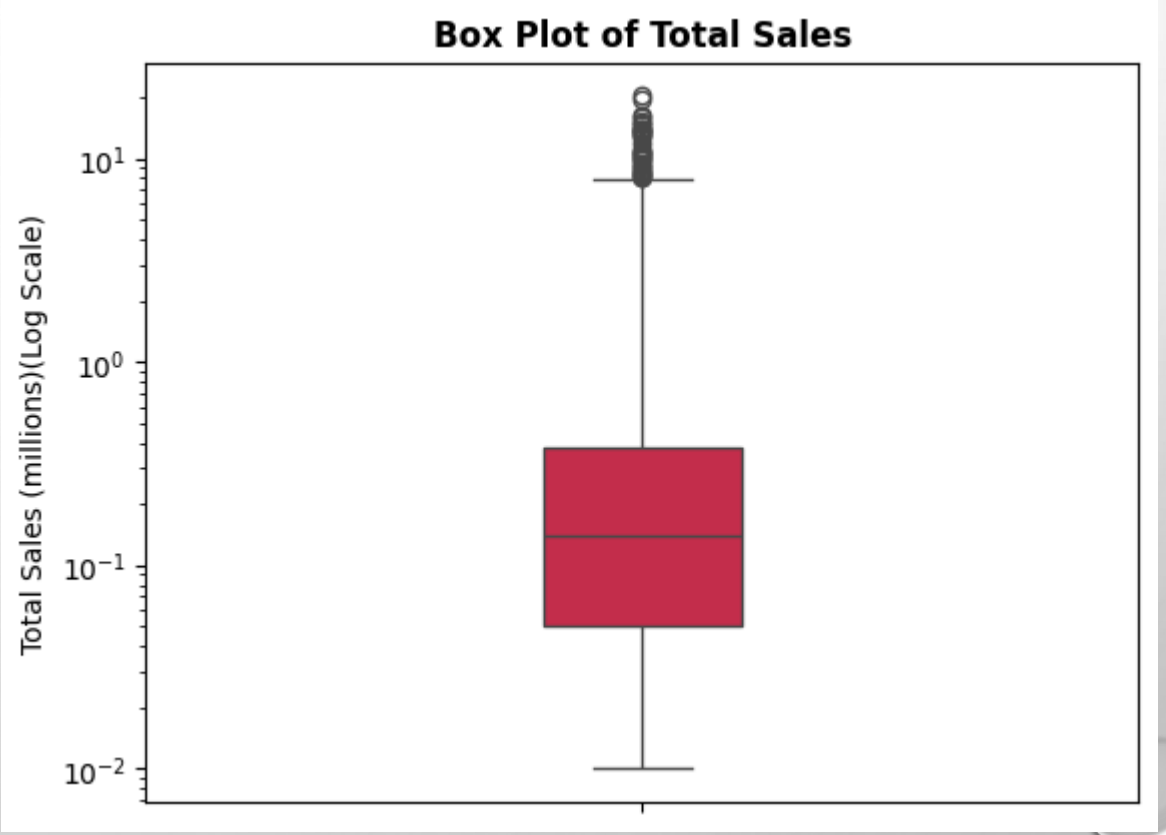


Dominated by GTA, COD, RDR Franchise

Univariate Analysis

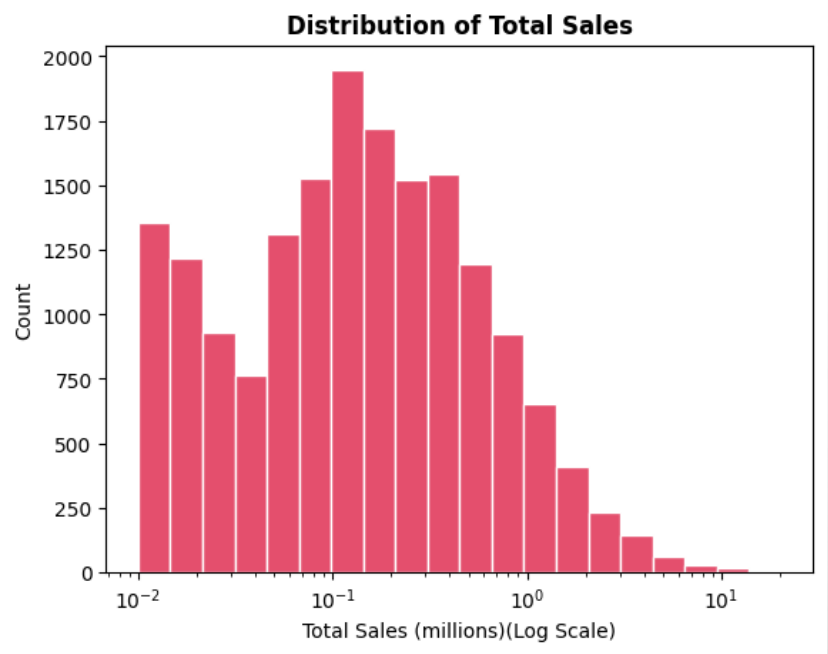
- Distribution of total sales (Histogram + Box plot)
- Log scale is applied to total sales for clearer visualization

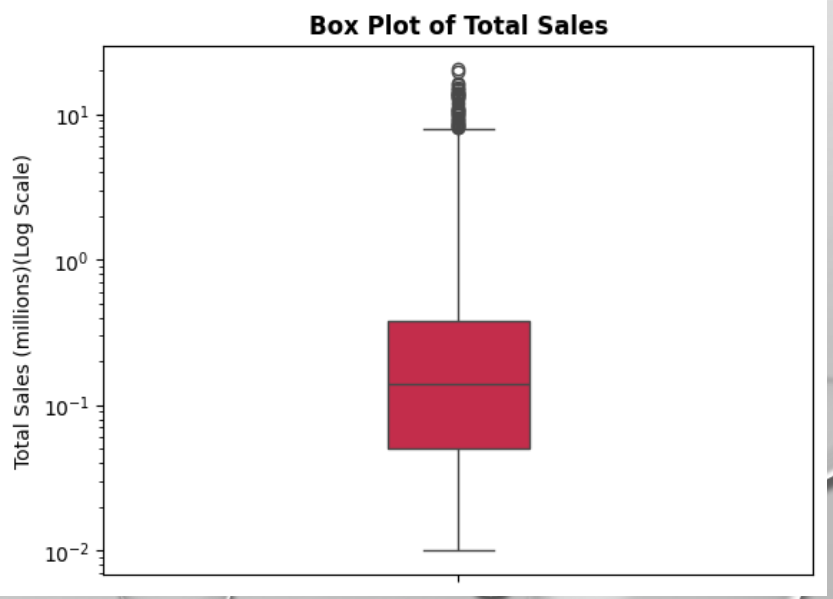




Observations and insights

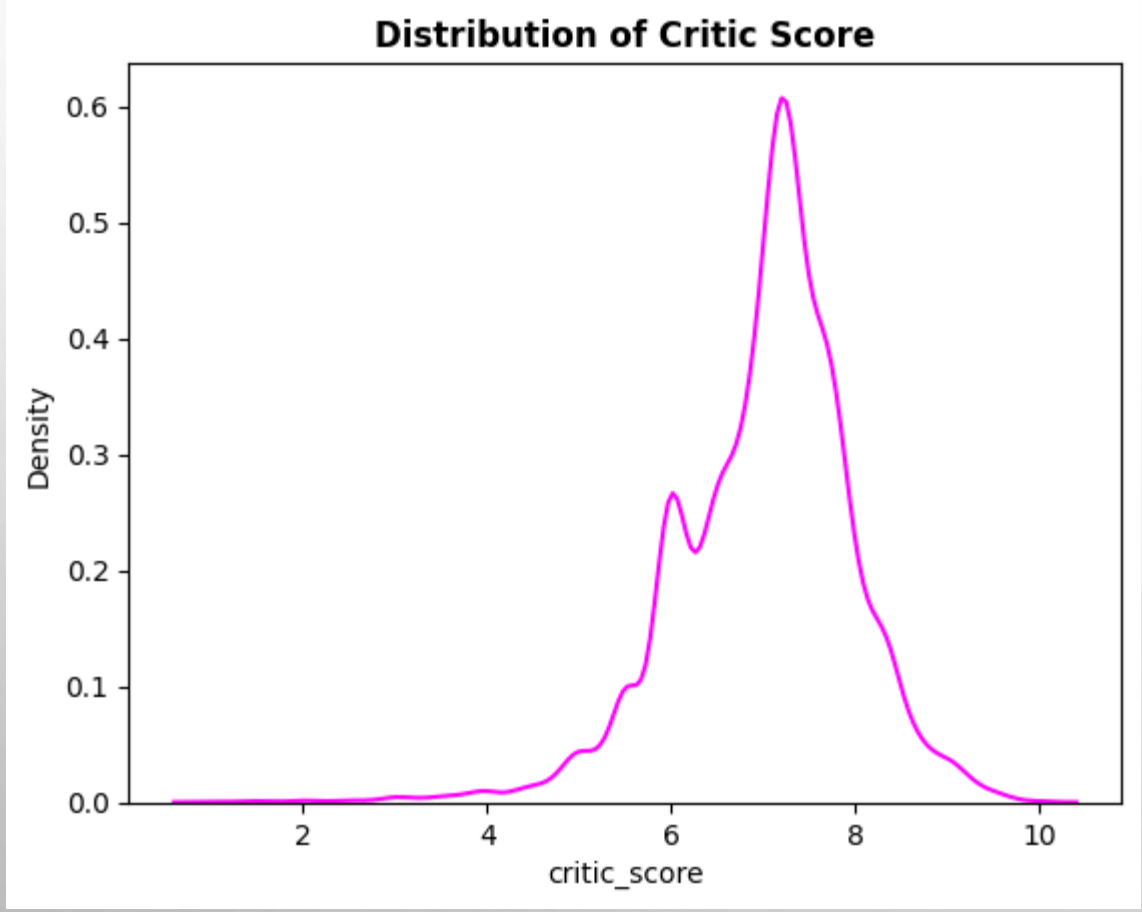
- Right-skewed distribution
- Many outliers
  - Most games have modest sales
  - Top sales are dominated by few games like GTA, COD and FIFA
- Major publishers should continue these IPs rather than developing a new one to mitigate financial risk

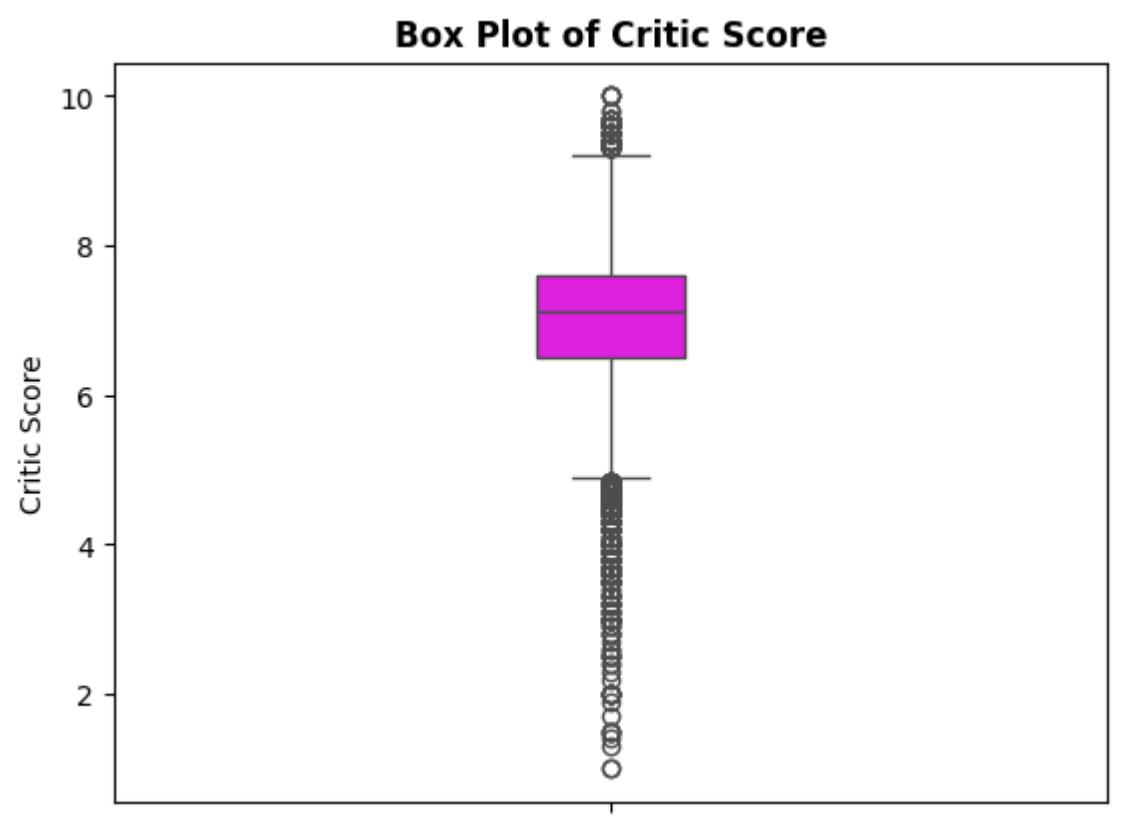




Univariate Analysis

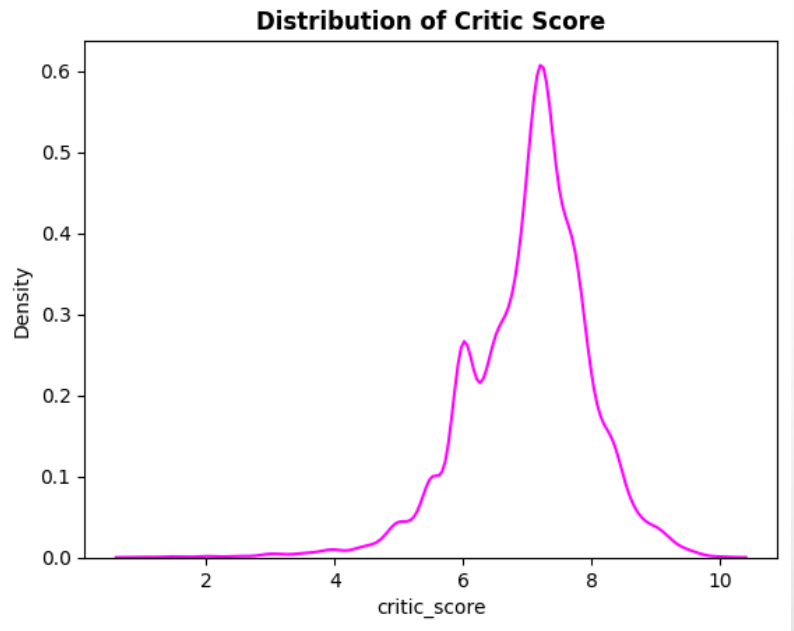
- Distribution of critic score (KDE plot + Box plot)

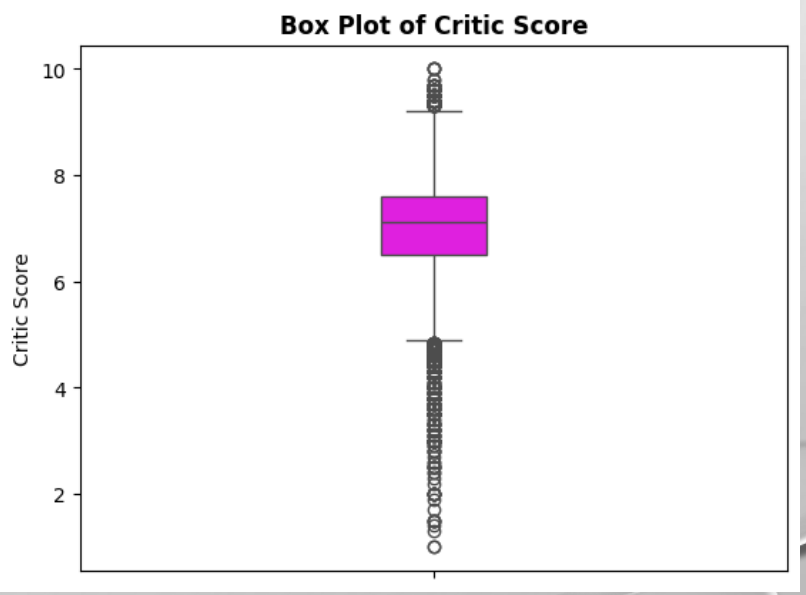




Observations and insights

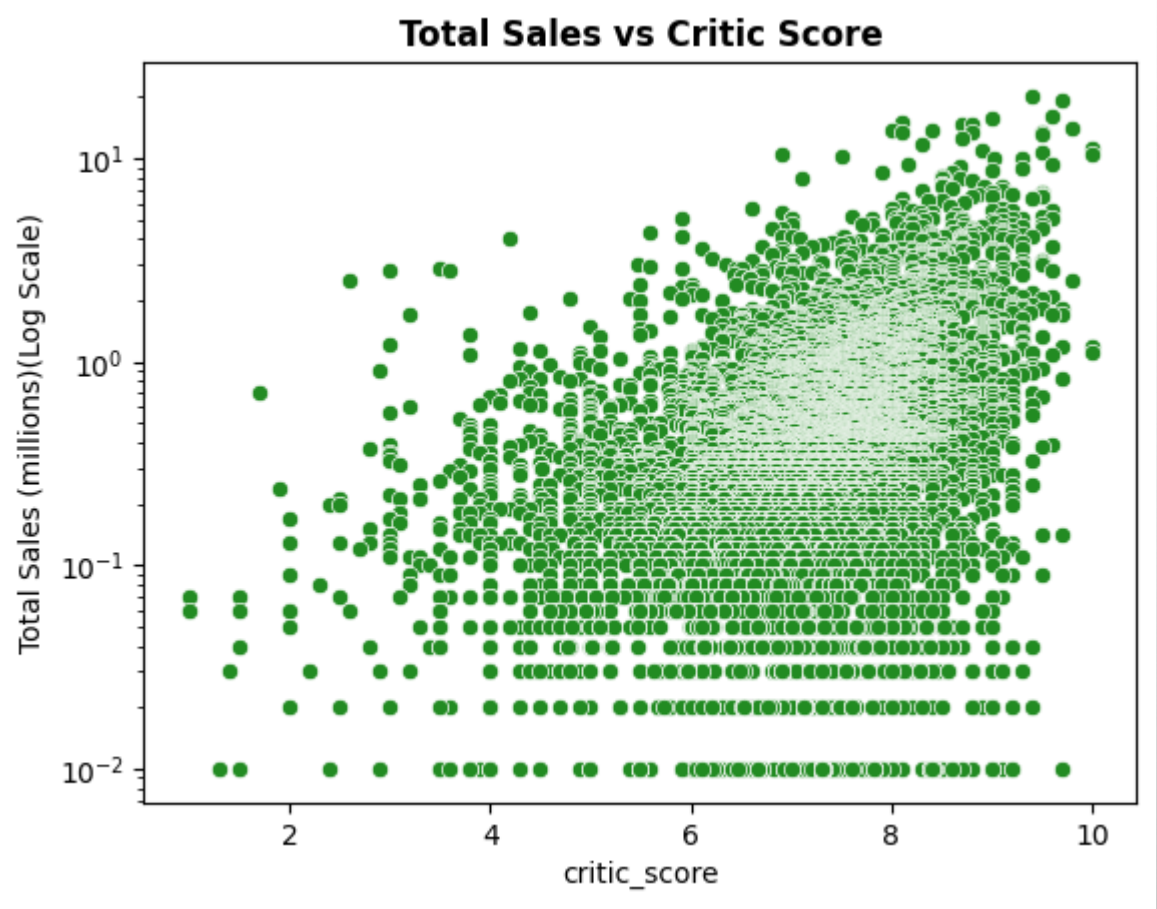
- Left-skewed distribution
- Median ~ 7, mode ~ 6.7
- Bilateral outliers
  - Scores are extreme
- Industry standard of 7 should be achieved by developers to avoid negative perception





Bivariate Analysis

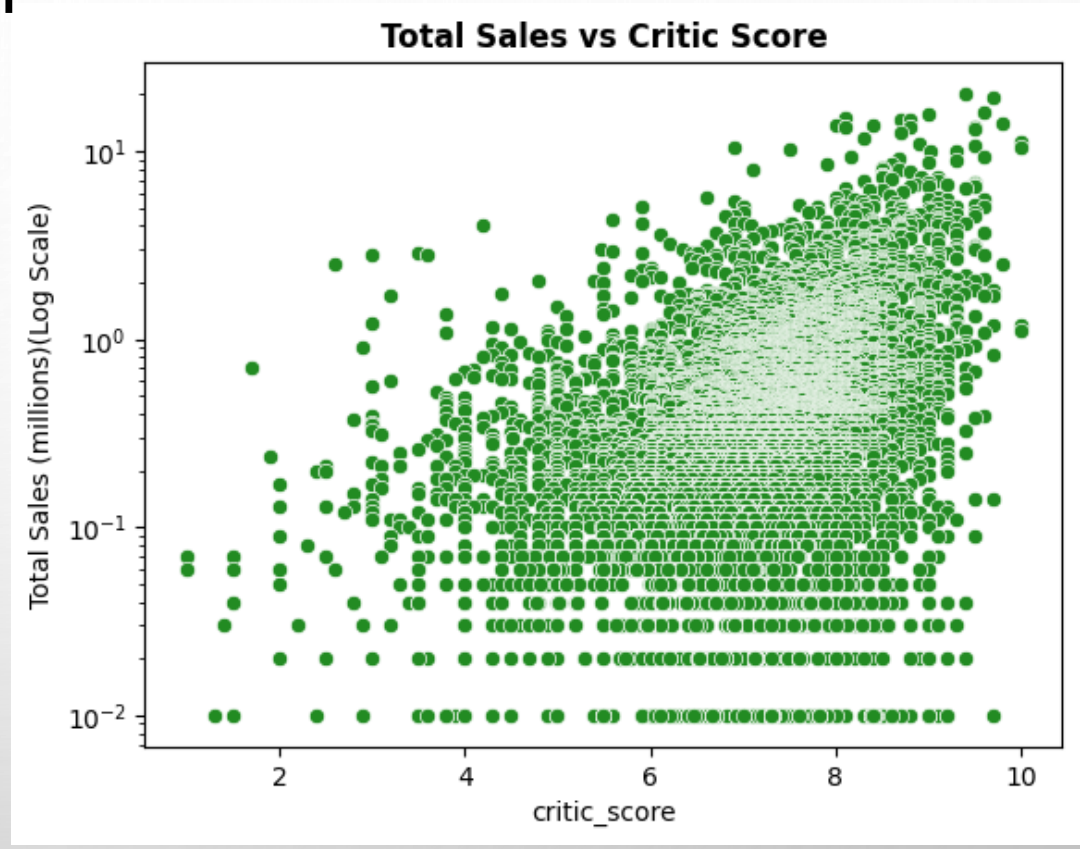
- Total sales vs critic score (Scatter plot)
- Question: Must higher critic score indicate better sales?



Correlation coefficient r = 0.33695185965566454

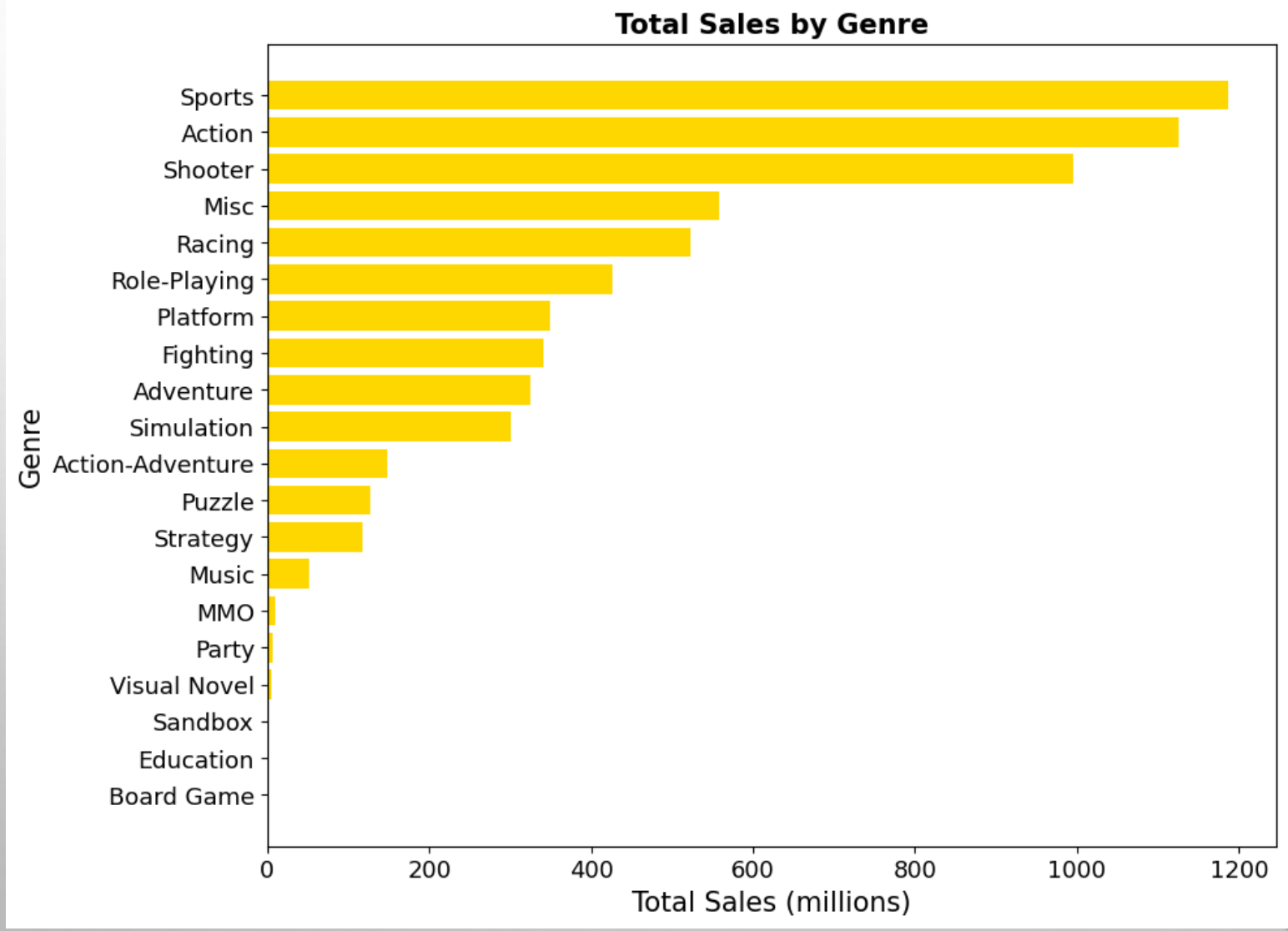
Observations and insights

- r = 0.337, a positive but weak relationship
  - Score alone is not an ultimate predictor of sales
- Within score of 5 – 9, sales have significant variance
  - Sales is affected by external factors like marketing spend, brand loyalty and release timing
- Below score of 3, there are rarely "breakout hits"
  - Moderate to high scores is not a guarantee of success, but a prerequisite of it



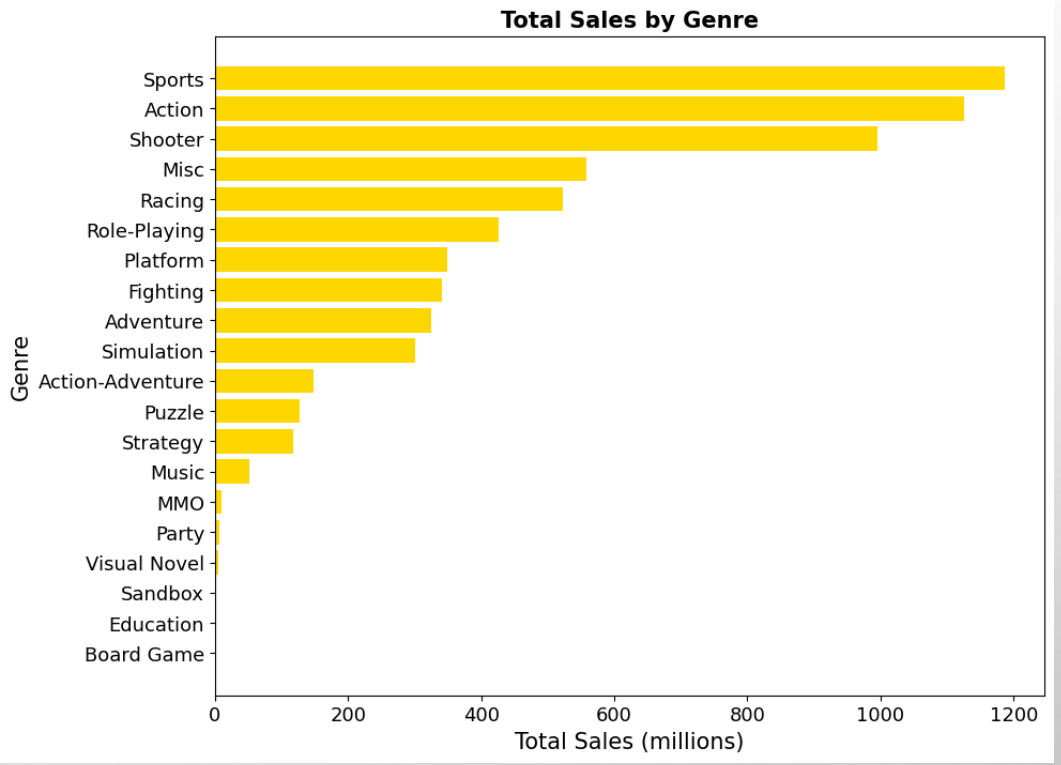
Bivariate Analysis

- Gamers' preference by genre (horizontal bar chart)



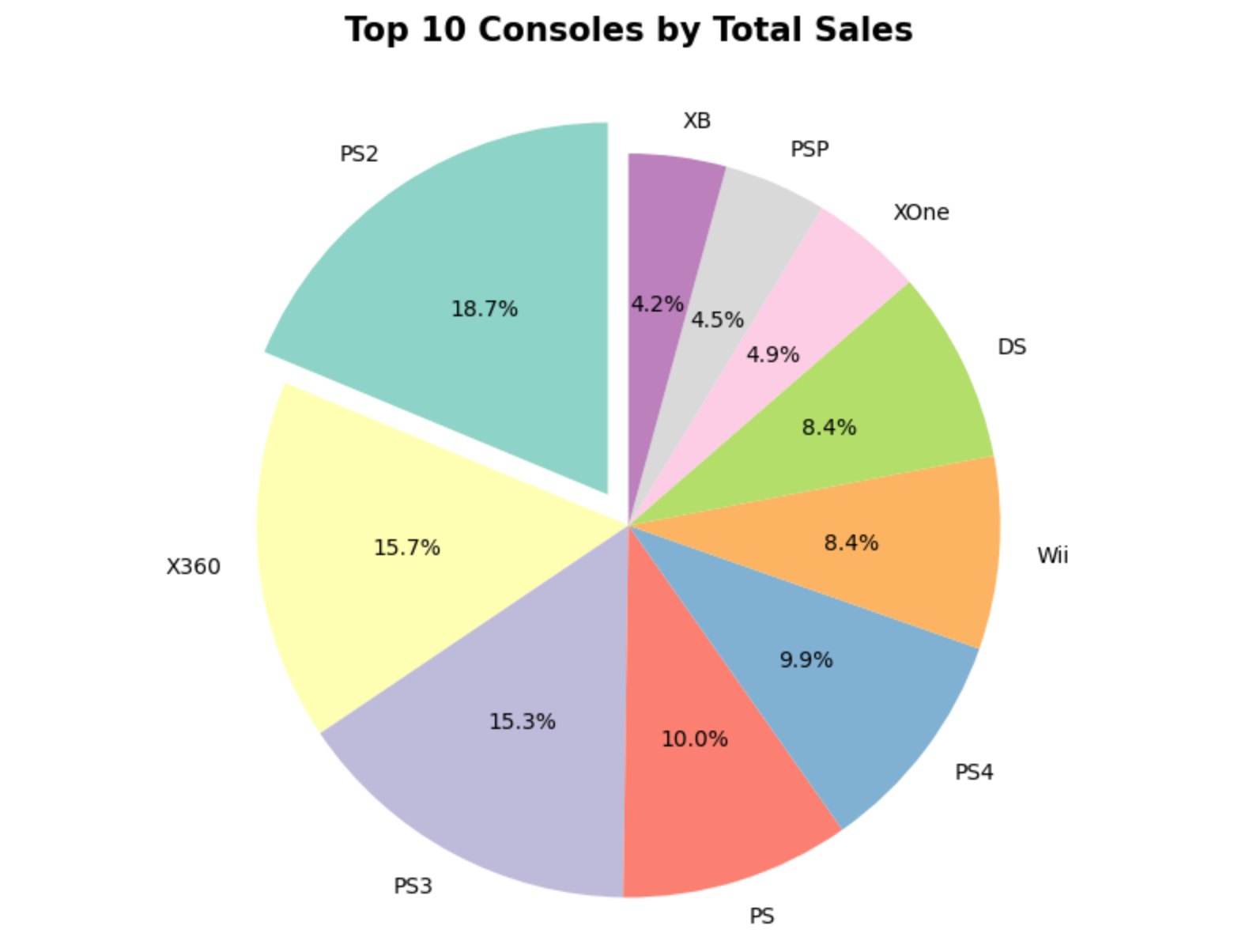
Observations and insights

- Top 3 genres: Sports, Action and Shooter
  - Developers should prioritize these genres to maximize profits
- Lower-budget genres like Simulation, Puzzle and Visual Novel have relatively lower sales
  - Genre is a more critical factor of success than score
  - Mediocre Action >> High-quality Visual Novel in terms of sales



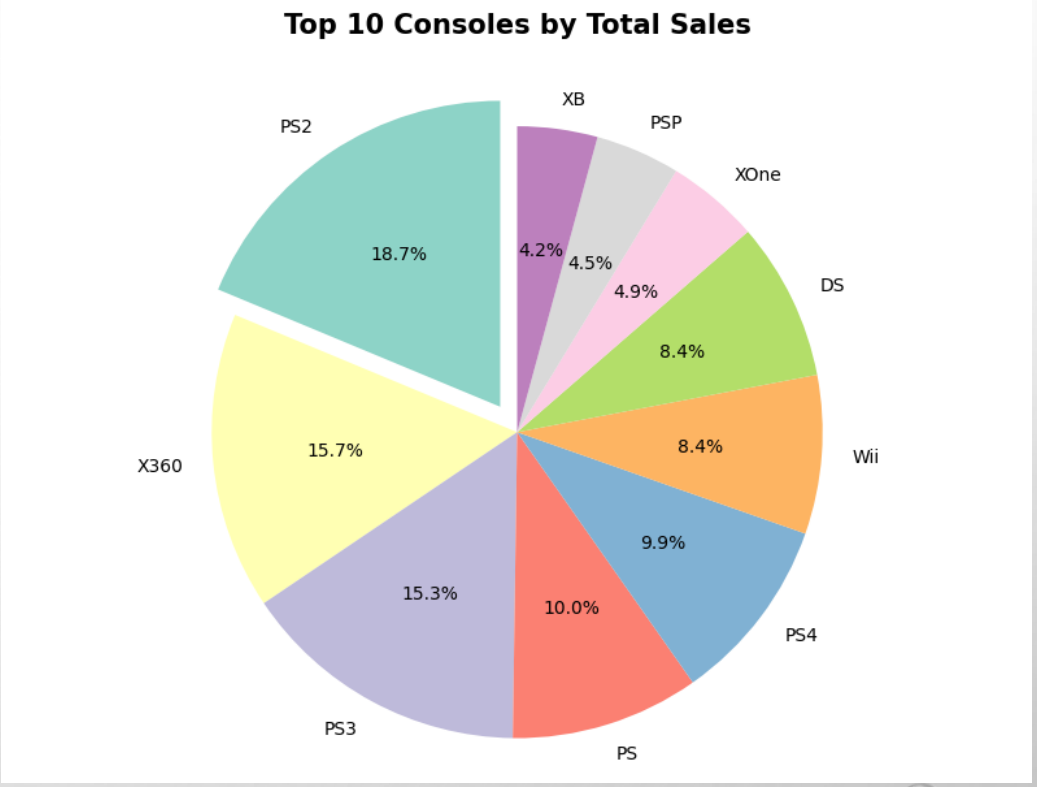
Bivariate Analysis

- Market share by consoles (pie chart)



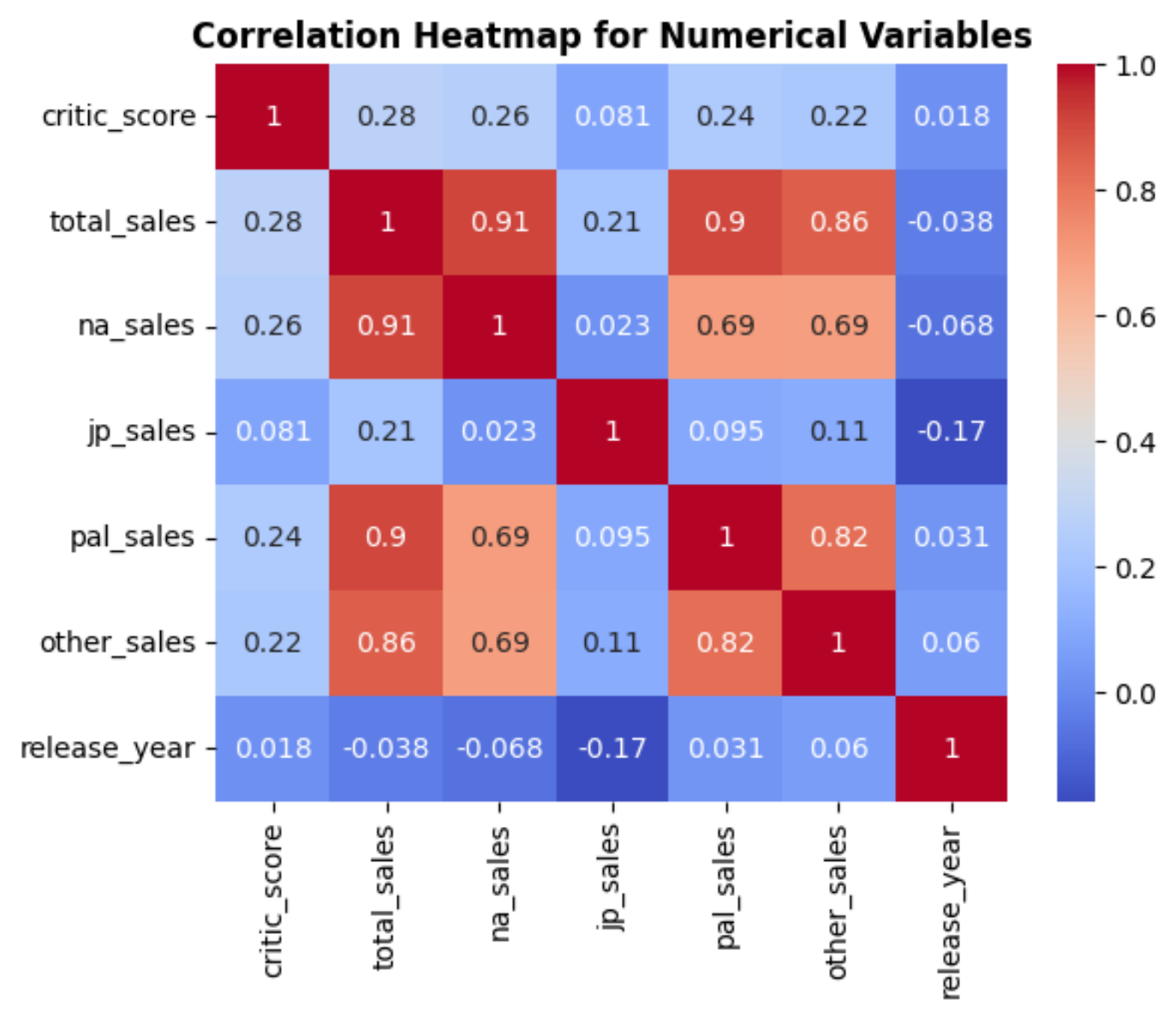
Observations and insights

- Sony has the largest console market in history (PS2+PS3+PS+PS4+PSP)
  - "Generation carry-over"
- Xbox360 and PS3 have nearly the same market share
  - Success of Microsoft is driven by Xbox Live online
- Publishers should avoid platform exclusivity
  - Many PlayStation-exclusive games port to PC few years later
- Handheld markets cannot be neglected
  - Former handheld developers should now target mobile market



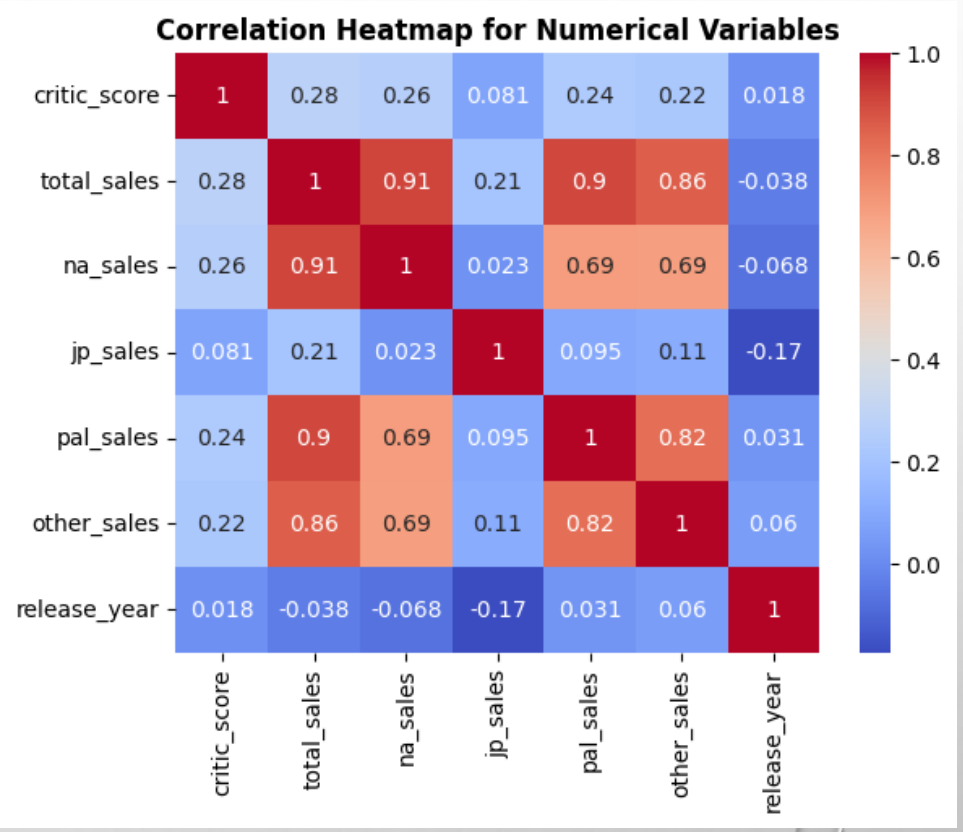
Multivariate Analysis

- Correlation heatmap (critic score, total sales, regional sales, release year)



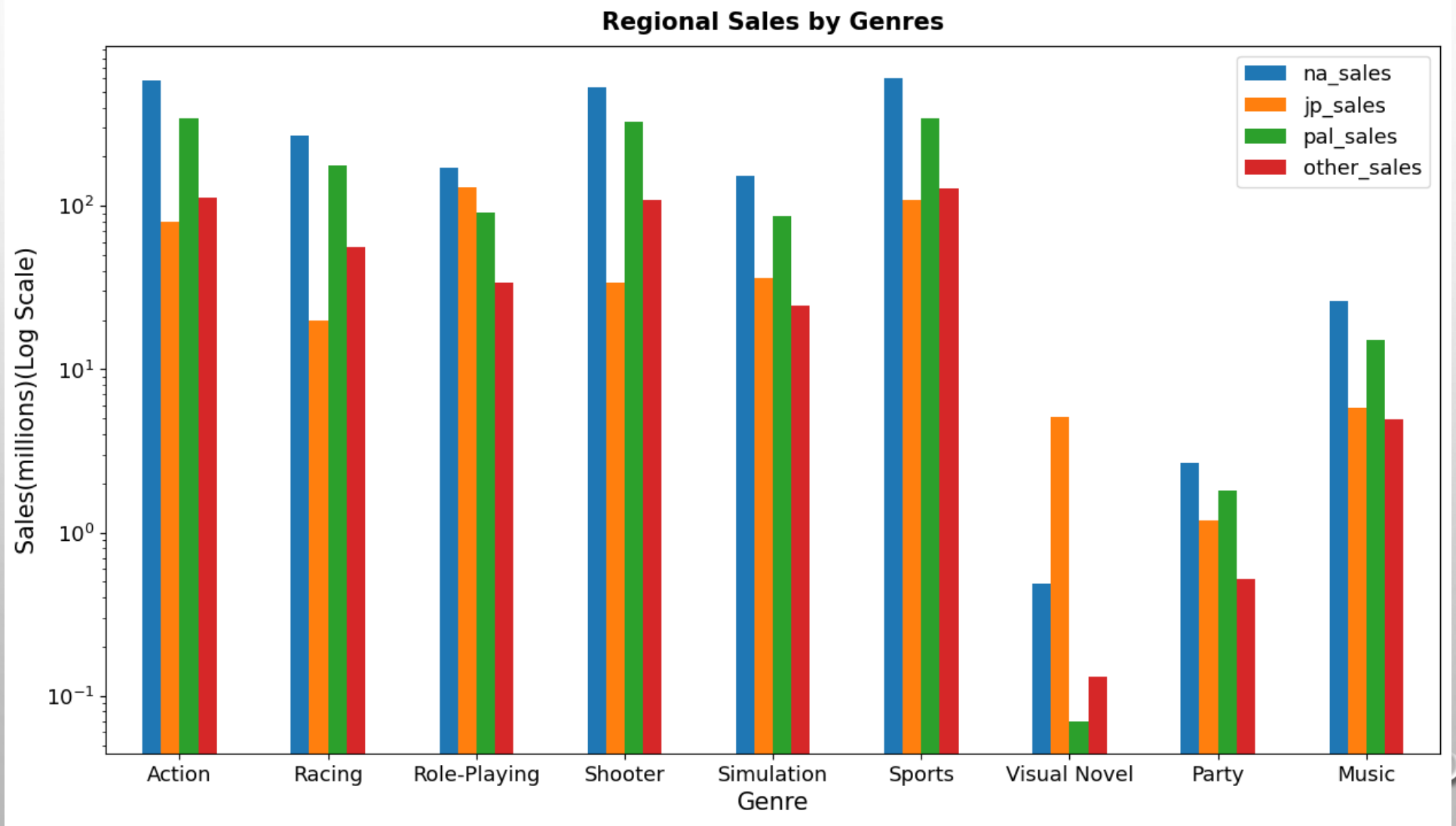
Observations and insights

- Global sales has high correlation with North American sales (0.91), European sales (0.9) and sales in other regions (0.86), except for Japanese sales (0.21)
  - Japan is a distinct and independent market
  - Cultural adaptation for Japanese market (Special character design, themes)
  - Pre-order sales in NA used as an indicator for marketing spend in Western countries
- Developers should invest heavily on graphic and storytelling if they want to target Western markets



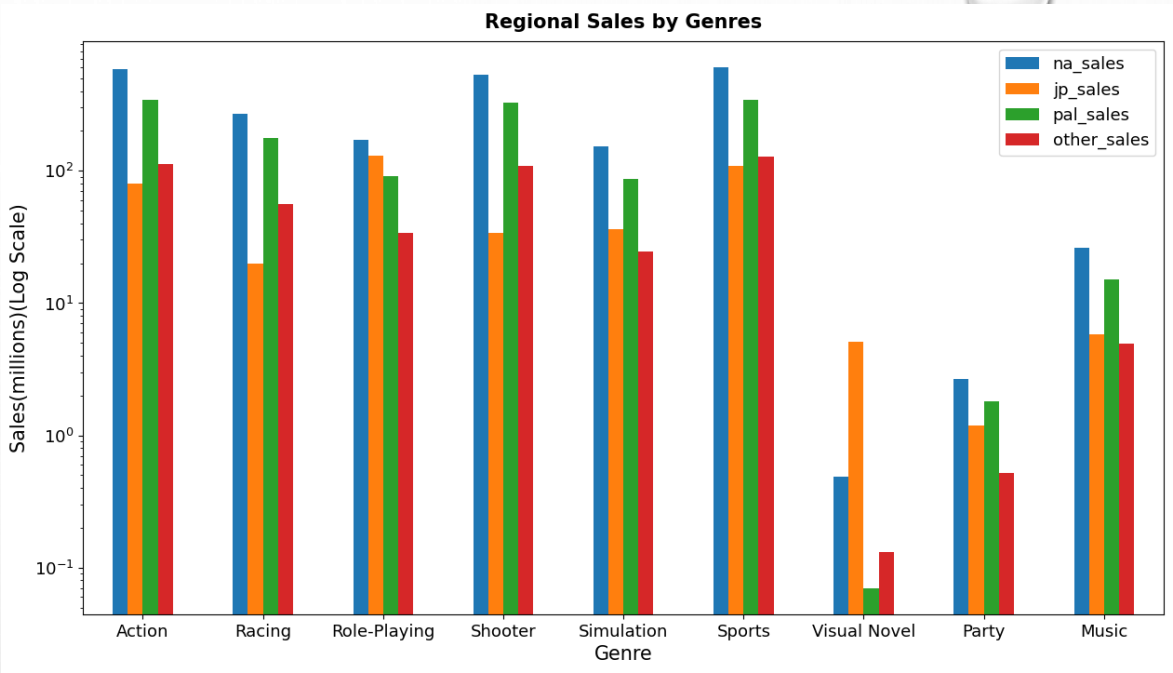
Multivariate Analysis

- Regional sales by genres (bar chart)



Observations and insights

- Action, Shooter and Sports games perform well globally except for Japan
  - Publishers of these games should prioritize digital distribution in Japan to reduce cost
- Japan has unique preference for RPG and Visual Novel games
  - Entry point for indie developers due to low cost



Challenges

- "One Man Army"
  - Coding, report and presentation all by myself
- Handling missing values and zeros
  - Remove them all would remain ~1700 rows (not enough and not accurate)
  - Handle each column one by one
- Handling outliers
  - "total_sales": values between 0.01 and 20
  - Plotting "total_sales" results in poor visualization
  - Apply logarithm to "total_sales" for clearer visualization

Limitations of the dataset

- Dataset contains digital sales? (Steam, Epic, PS Store)
  - If not, sales number is greatly underestimated
- Dataset does not include cancelled games

Future improvement

- Compare pre-release data (no. of likes from social media, trailer view counts in YouTube, Steam "Wishlist") with final sales
  - Are these data a better predictor of sales than critic score?
- How subscription service (Xbox Game Pass, PS Plus, EA Play) affects the sales of a game?
  - Popularity reflected by total sales or no. of downloads in subscription service?

Ideas that were not implemented and the reasons

- Principal Component Analysis (PCA)
  - Find the principal components that represent variables of similar attributes
  - Reduce the number of variables for easy analysis
  - Complicated and time-consuming
- Random Forest
  - Consists of many decision trees
  - Each decision tree contains criteria used to predict the results
  - Might be more accurate than KNN Imputation
- Time Series Forecast
  - Could be used to predict future sales
  - Not implemented because it is not accurate to predict future sales using past data

Summary and key insights

- Franchise Continuity: continue current IPs instead of developing a new one
- Quality Standard: achieve a minimum of score 7 to sell millions of copies
- Genre-First Development: target popular genres rather than niche genres
- Cross-Platform Strategy: multi-platform releases rather than platform-exclusive
- Mobile Market: former handheld developers should now target mobile market
- Localization of Japanese Market: special character designs and add RPG elements to penetrate Japanese market
- Subscription Service: small studios should join subscription service to enhance sales and exposure

Reference

Divekar, B. (2024). Video Game Sales and Industry Data (1980-2024) [Data set]. Kaggle. https://www.kaggle.com/datasets/bhushandivekar/video-game-sales-and-industry-data-1980-2024

The End